



Nanoindentation mapping of the mechanical properties of human molar tooth enamel

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Abstract

The mechanical behavior of dental enamel has been the subject of many investigations. Initial studies assumed that it was a more or less homogeneous material with uniform mechanical properties. Now it is generally recognized that the mechanical response of enamel depends upon location, chemical composition, and prism orientation. This study used nanoindentation to map out the properties of dental enamel over the axial cross-section of a maxillary second molar (M^2). Local variations in mechanical characteristics were correlated with changes in chemical content and microstructure across the entire depth and span of a sample. Microprobe techniques were used to examine changes in chemical composition and scanning electron microscopy was used to examine the microstructure. The range of hardness (H) and Young's modulus (E) observed over an individual tooth was found to be far greater than previously reported. At the enamel surface $H > 6$ GPa and $E > 115$ GPa, while at the enamel–dentine junction $H < 3$ GPa and $E < 70$ GPa. These variations corresponded to the changes in chemistry, microstructure, and prism alignment but showed the strongest correlations with changes in the average chemistry of enamel. For example, the concentrations of the constituents of hydroxyapatite (P_2O_5 and CaO) were highest at the hard occlusal surface and decreased on moving toward the softer enamel–dentine junction. Na_2O and MgO showed the opposite trend. The mechanical properties of the enamel were also found to differ from the lingual to the buccal side of the molar. At the occlusal surface the enamel was harder and stiffer on the lingual side than on the buccal side. The interior enamel, however, was softer and more compliant on the lingual than on the buccal side, a variation that also correlated with differences in average chemistry and might be related to differences in function. © 2002 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Enamel is the hardest biological substance in the human body and is a composite material consisting of both a mineral and an organic phase. The mineral phase predominates (95–96 wt.%) and consists primarily of calcium phosphate salts in the form of large hexagonal hydroxyapatite crystals that are both carbonated and defective. Sets of similarly

orientated crystals form rod-like structures called enamel prisms, 3–6 μm in cross-sectional diameter. Prisms are separated from each other by a thin organic prism sheath and by interprismatic enamel. The protein/organic matrix comprises approximately 1 wt.% of the enamel, and the remaining approximately 3% is contributed by water (Williams et al., 1989). Knowledge of the mechanical characteristics of dental enamel, and the relation between these properties and its chemistry and microstructure, could provide valuable insights into, for example the study of tooth wear and the development of improved toothpastes, oral treatments and dental restoratives.

Mechanical studies of tooth enamel have often viewed it as a homogeneous solid (e.g. Remizov et al., 1991; Willems et al., 1993). Other investigations, however, have begun to explore variations in the mechanical properties of enamel

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that relate to test location, local chemistry and prism orientation. For example, Knoop microhardness tests (Craig and Peyton, 1958; Meredith et al., 1996) and compression tests (Craig et al., 1961) have suggested that the Young's modulus (E) and hardness (H) may be slightly higher for cusp (or surface) enamel than for side (or sub-surface) enamel, while investigations with a depth-sensing Vickers indenter (Xu et al., 1998) have shown that the H and E obtained from an occlusal section of enamel are generally higher than those obtained from an axial section.

In an effort to explain variations in the mechanical properties of enamel with location, researchers have hypothesized that local chemistry, i.e. degree of tissue mineralization, could be a critical factor. Notably, correlations between microhardness and mineral content have been found in sound human enamel (Kodaka et al., 1992), and E was estimated to be capable of changing by 3 GPa for each 1 vol% change in mineralized hydroxyapatite (Staines et al., 1981). In fact, finite-element modeling of the E of enamel has been conducted, using the volumetric crystalline fraction of hydroxyapatite to calculate the possible ranges of values for stiffness, both parallel and perpendicular to crystal orientation (Spears, 1997): parallel to the crystal direction, it was found that E increases from 93 to 113 GPa as the crystalline fraction increases from 0.81 to 0.99; perpendicular to the crystal direction, E shows a much larger variation, from 19 to 91 GPa, over the same range of crystalline fraction.

In addition to having variations in mineral content, hydroxyapatite is known to be an elastically anisotropic material and spherical-indentation testing has indicated that differences in mechanical properties may be due to the variations in prism orientation. Staines et al. (1981) found that enamel hardness was higher when the indentation direction was perpendicular to the c -axis of the hexagonal hydroxyapatite crystals (parallel to the basal planes), and lower when the indentation direction was coaxial. In a similar manner, Rasmussen et al. (1976) found that enamel's work of fracture is very anisotropic, being lowest parallel to the prisms. Twenty years later, Xu et al. (1998) report that the fracture toughness of enamel can differ by a factor of three, depending on the alignment of prisms relative to the orientation of the crack.

These mechanical characterizations strongly suggest that one should consider intra-tooth variations in the properties of enamel when developing dental restorations or when comparing teeth from different populations, species, tooth types, etc. Our goal was to expand and elaborate upon the idea of enamel heterogeneity within individual teeth by determining and mapping local variations in mechanical properties across the entire depth and span of a human enamel sample.

We chose nanoindentation to determine H and E at multiple locations within individual enamel samples, because it can accurately measure the mechanical properties of very small, submicrometre, volumes with fine spatial resolution. The Nanoindenter XPTM used here has a lateral-positioning resolution of 1 μm and can apply loads of just a few μN s

while simultaneously measuring displacement changes <1 nm. Traditional methods for determining H only take into account permanent (plastic) deformation and require the visualization and measurement of indents (e.g. Craig and Peyton, 1958; Meredith et al., 1996). Nanoindentation, however, accounts for both plastic and elastic deformation, as well as time-dependent effects. In addition, it removes the requirement of indent visualization because both depth and applied load are continuously monitored throughout the indentation process (Oliver and Pharr, 1992).

Nanoindentation has been utilized previously for the examination of dental tissue. Kinney et al. (1996) used it to show that the H and E of intertubular dentine decrease with distance from the enamel–dentine junction. Similarly, van Meerbeek et al. (1993) used nanoindentation to characterize the H and elasticity of the resin–dentine bonding area. In the case of enamel, however, nanoindentation has generally only been used to determine 'average' values of H and E that are not site-specific: 3.39 ± 0.18 and 90.59 ± 16.13 GPa, respectively (Willems et al., 1993). Here, our goal was to use the spatial sensitivity of nanoindentation to quantify the H and E of enamel throughout its full volume in an individual tooth, thereby identifying any significant local variations.

2. Materials and methods

2.1. Sample preparation

Two maxillary second molars (M^2) and one maxillary third molar (M^3) were characterized in detail. Using a small number of teeth enabled a very large number of tests to be performed on each, thereby providing a detailed characterization of variations in mechanical properties, chemistry, and microstructure across the full depth of enamel. Before testing, each of the three teeth was embedded in blocks of either Ultra-Mount acrylic mounting compound or Epo-Kwick epoxy mounting compound (Buehler Ltd., Lake Bluff, IL). The blocks were cut with a diamond saw so that the teeth were divided into two halves, with the cut lying perpendicular to the buccolingual division line. The cut surfaces of each half were then polished to a 1 μm finish using a motorized polishing wheel with interchangeable silicon carbide grinding discs of diminishing grit size (e.g. 240, 320, 400, 600, 800 and 1200). Diamond paste or alumina suspensions with decreasing grit size (9, 6, 3, and 1 μm) were used for the final polishing. Between each grit size and at the end of polishing, the sample surface was gently cleansed of debris using running water, a mild soap solution and cotton-tipped applicators. Ultrasonic cleaning was avoided because it tended to crack the samples and loosen their mounting.

2.2. Mechanical characterizations

The mesial halves of the prepared samples were chosen for nanoindentation testing, which was conducted along

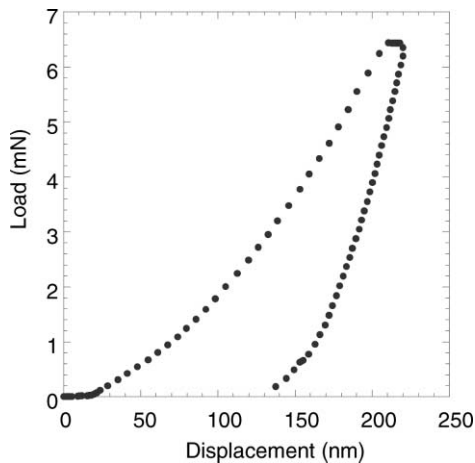


Fig. 1. Typical load vs. displacement curve for a nanoindentation in tooth enamel. The contact depth, area and stiffness were all determined at the maximum load by fitting a polynomial expression to the upper 70% of the unloading curve and were used to quantify H and E for each nanoindentation according to standard routines (Oliver and Pharr, 1992). A Poisson's ratio (ν) of 0.25 was assumed when calculating E from the reduced modulus, $E/1 - \nu^2$.

the outer perimeter of the enamel (the occlusal surface) and in arrays directed from the outer perimeter towards the enamel–dentine junction. An exemplar load–displacement curve generated by the Nanoindenter (MTS Systems Corp., NanoInstruments Innovation Center, Oak Ridge, TN) is shown in Fig. 1. Indents were typically made to a depth of 400 or 800 nm and the diamond tip used for testing had a Berkovich (three-sided pyramidal) geometry. The data were analyzed using standard routines (Oliver and Pharr, 1992) to provide H and E for each nanoindentation. The depth of each indentation was determined by fitting a polynomial expression to the upper 70% of the unloading curve. The contact area was then determined, using the depth to area calibration for the tip. H was calculated using the maximum load and the contact area at maximum load, E was calculated using the contact stiffness, the contact area at maximum load, and a value of 0.25 for Poisson's ratio.

Because the nanoindentation experiments were conducted in an ambient environment over a period of many hours, samples were stored in the same ambient environment to avoid excessive drying during testing. In this environment, the absolute values of the measured elastic moduli were expected to be slightly higher than they would be if the samples were tested in a wet environment, based on earlier studies (Staines et al., 1981) in which E increased by approximately 15% when a wet sample was allowed to dry in air for over 72 h. The same may be true for the H as well. However, this enhancement was expected to be fairly uniform over the entire enamel sample and, therefore, not to alter any distinct trends.

For each location on the enamel surface, values of H and E were determined using an average of 5–10 closely spaced

indentations. In total, 2000–3000 nanoindentations were performed across the entire enamel surface shown in Fig. 2. The spacings between individual indentations were typically 10–15 μm in the area where the mechanical properties changed sharply (e.g. near the occlusal surface) and larger (>20 μm) in regions where the properties varied gradually.

2.3. Chemical and microstructural characterizations

Chemistry was studied by electron-microprobe analysis and structural information was obtained by scanning electron microscopy. The microprobe analysis was performed on a JEOL 8600 Super Probe (JEOL, Danvers, MA) operating with a 20 nA beam current, 15 keV accelerating potential, 60 s counting times, and a beam of 10 μm diameter. Two sets of chemical data were obtained. One measured the elements Na, Mg, Al, Si, P, K, Ca, Ti, Cr, Mn, Fe, Y, Zr, Ba, La, Ce, Pr, Nd, Sm, Gd, Dy, Er, Yb, Hf, Ta, Pb, Th, and U. With the exception of Na, Mg, Al, P, K, Ca, and Ti most elements were below detection limits ($\sim 0.1\%$). The second set included the elements F, Na, Mg, Al, Si, P, Cl, K, Ca, Ti, Cr, Mn, and Fe. Standards used to calibrate the microprobe included natural and synthetic minerals.

3. Results

3.1. Mechanical characterizations

The results of the nanoindentation tests for a maxillary M^2 are shown in Fig. 2 as a map of mechanical properties throughout the enamel. Matlab (version 5.2) was used to interpolate between the individual values of H and E , thereby creating a continuous map. The S.D.s for these averages ranged from 0.2 to 0.3 GPa for H and from 2 to 5 GPa for E , which was $< \pm 10\%$ in both cases. Also included in Fig. 2 are previous reports of average values for the H and E of enamel. As these previous studies utilized different measurement techniques, the absolute values may depend slightly on which technique was used. However, their inclusion here provides an important comparison between the previous averages and the actual variations that can be seen in Fig. 2.

The maps of mechanical properties (Fig. 2a and b) shows that H and E decreased 'on average' from 4.6 to 3.4 GPa (a 26% decrease) and from 91.1 to 66.2 GPa (a 27% decrease), respectively, on going from the enamel surface to the enamel–dentine junction. The extreme variation, however, from the maximum H of 6.4 GPa to the minimum H of 2.7 GPa was well over 50%. Equally large variations were observed in E (120–47 GPa).

Note in Fig. 2a and b that there was a significant drop in H (>50%) and E (>50%) on moving from the enamel surface to the enamel–dentine junction, for both the buccal and lingual sides of the molar. Note also that the values of H and E for the interior of the lingual side of the molar were generally lower than those found in the interior of the

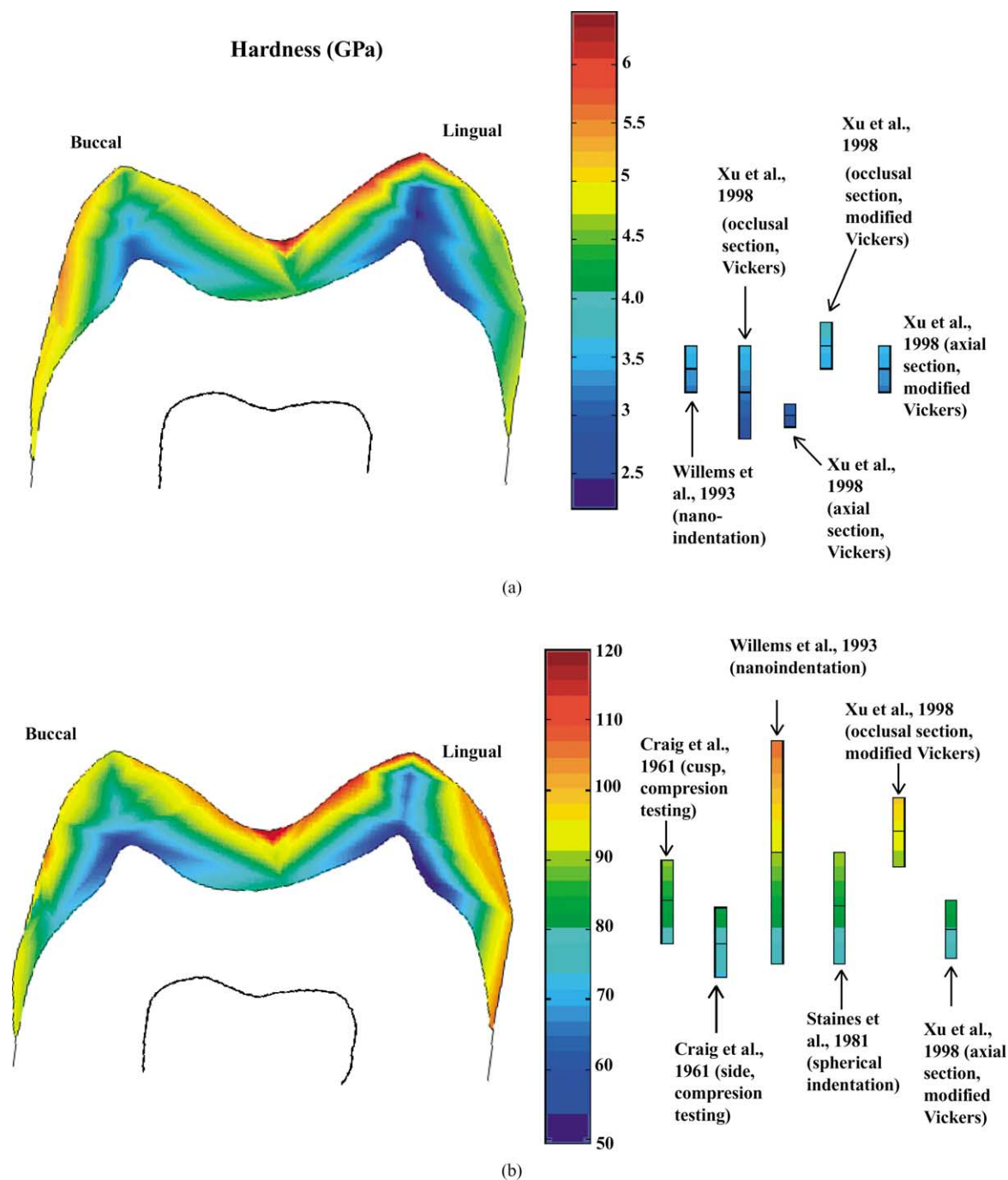


Fig. 2. Enamel H (a) and E (b) results for the mesial half of a maxillary M^2 as determined by nanoindentation. The S.D.s for these averages range from 0.2 to 0.3 GPa for H and from 2 to 5 GPa for E . Note the wide variation in mechanical properties between the enamel surface and the enamel–dentine junction. Average values of H and E that have been reported earlier by other researchers are included for comparison.

buccal side. The maximum H (>6 GPa) and E (>110 GPa), however, appeared at the lingual cusp and extended along its occlusal surface to the centre of the molar. In contrast, the occlusal surface of the buccal cusp had a relatively low H (4.6 GPa) and E (93 GPa).

To determine whether the intra-tooth variations reported in Fig. 2a and b are common to other molars, a second maxillary M^2 and a maxillary M^3 (each from different individuals) were also examined. The trends in H and E observed in the second maxillary M^2 were similar to those seen in the first maxillary M^2 , as shown in Fig. 6, suggesting that the properties of M^2 could be similar for different individuals. Comparing the properties of the M^2 and M^3 , the trends in H and E suggest these molars have similar surface properties but different interior properties. The interior regions below the buccal and lingual cusps of M^3 were generally harder and stiffer than the corresponding regions in M^2 , particularly for the lingual cusp (Fig. 6).

3.2. Chemical characterizations

Compositional variations in the distal half of the M^2 shown in Fig. 2 were characterized using electron-microprobe analysis to examine the role of chemistry in determining the mechanical behavior of the enamel. The sample was first coated with a thin layer of carbon (approximately 5 nm-thick) using a carbon evaporator to provide a conductive sample surface. Points of analysis were located within three different areas: at the centre of the cross-sectional face, in a region initiating at the lingual cusp and in a region

Table 1
Representative electron-microprobe analyses (%)

	Enamel (rim)	Enamel (core)	Dentine
F	0.030	0.055	0.106
Na ₂ O	0.585	1.234	0.726
MgO	0.234	0.570	1.473
Al ₂ O ₃	0.008	0.007	0.007
P ₂ O ₅	40.96	38.34	34.99
Cl	0.592	0.093	0.086
K ₂ O	0.037	0.073	0.027
CaO	51.73	47.98	43.83
Total	94.18	88.35	81.25

beginning at the buccal cusp. Within each area, microprobe tests were conducted at points along a line perpendicular to the occlusal surface. The lines traversed from a starting point close to the occlusal surface to a corresponding point within the dentine. Sites for rare earth-element analysis were spaced 100 μ m apart, and sites for fluorine and chlorine measurements were spaced 50 μ m apart.

Selected analyses of the distal half of the molar in Fig. 2 are presented in Fig. 3 and Table 1. In the previous studies, the major inorganic compounds of human tooth enamel were found to be calcium, phosphate, carbonate, magnesium, and sodium (Bodart et al., 1981). In the present analysis, the wt.% of P₂O₅ and CaO decreased on moving from the outer rim of the enamel towards the dentine. This was true for all three traverses: centre, buccal, and lingual. MgO, Na₂O, and

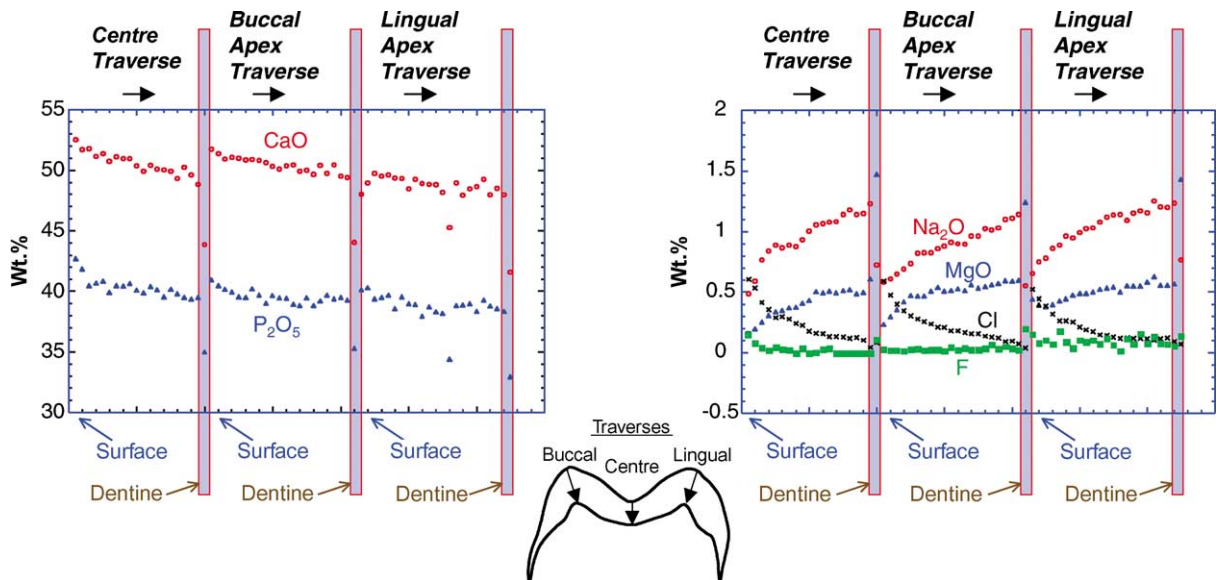


Fig. 3. Results of the electron-microprobe chemical analysis of the distal half of the molar shown in Fig. 2. The values are wt.% as a function of location. The plots are each broken into three sections for the three different traverses shown in the schematic. Points on the left-hand side of each section represent values at the enamel surface, points on the right-hand side correspond to interior values near the enamel–dentine junction. The points within the grey, vertical bars represent values found in the dentine.

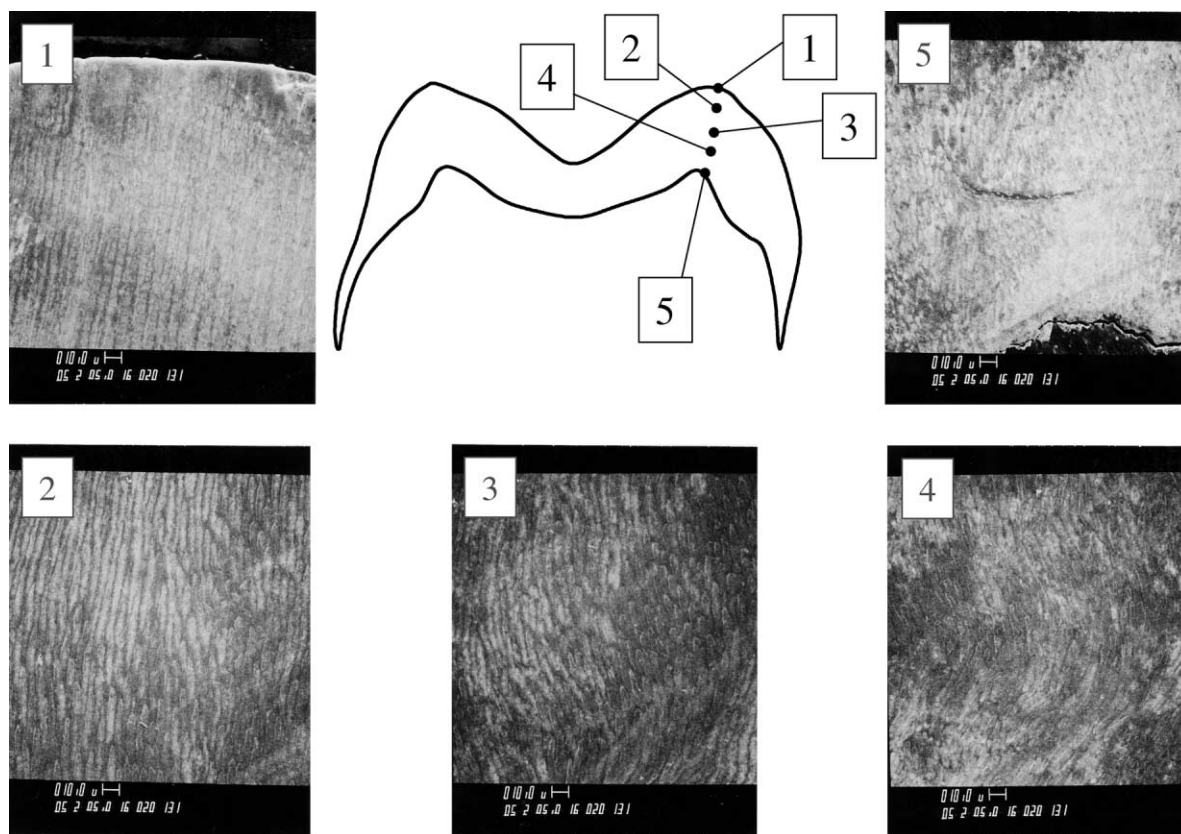


Fig. 4. Lingual traverse: scanning electron microscopic images of the axial section surface of a maxillary M².

K₂O (not shown) displayed the opposite trend with significant increases on moving towards the enamel–dentine junction. Fluorine content, though, showed no significant statistical trend, though the centre traverse showed a slightly higher fluorine content at the enamel surface.

Dramatic chemistry changes at the “enamel/dentine” interface were also visible from these results. Going from enamel to dentine, MgO showed a 134% increase (averaged over the three traverses), and Na₂O, CaO, and P₂O₅ underwent 43, 11, and 12% decreases, respectively. Going from enamel to dentine, the fluorine content was found to increase by over 100% in each traverse.

3.3. Microstructural characterizations

To investigate the effects of the enamel’s microstructure on the observed mechanical properties, scanning electron microscopy was used to examine the mesial half of the same M² (after the mesial half was used to obtain the mappings of Fig. 2). The specimen was first etched for 15 min with an aqueous 0.005 M citric acid solution, then mounted on a conductive metal stub and sputter-coated with gold–palladium. Imaging was done at a 5 kV accelerating voltage with a secondary electron detector at 500×

magnification. Exemplar photos from scanned imaging are shown in Figs. 4 and 5. Fig. 4 shows the change in structure from the lingual cusp to the enamel–dentine junction and Fig. 5 shows the change in structure from the buccal cusp to the enamel–dentine junction.

At the lingual cusp the enamel prisms were normal to the occlusal surface (see Fig. 4) but their orientation was less well defined in the interior of the enamel. The prisms lay in a number of different directions close to the enamel–dentine junction, with many of the prisms at angles <90° to the occlusal surface. The prisms at the buccal cusp were mostly normal to the occlusal surface, but not as clearly ordered as the prisms at the lingual cusp. The interior prisms on the buccal side were generally not normal to the surface, and this was most obvious at the enamel–dentine junction.

4. Discussion

The maps of mechanical properties in Fig. 2 and the plots in Fig. 6 show significant decreases in *H* and *E* on going from the enamel surface to the junction with dentine, particularly for the lingual side. The extreme variations are well over 50% for both *H* and *E*, and far exceed the S.D.s (<±10%)

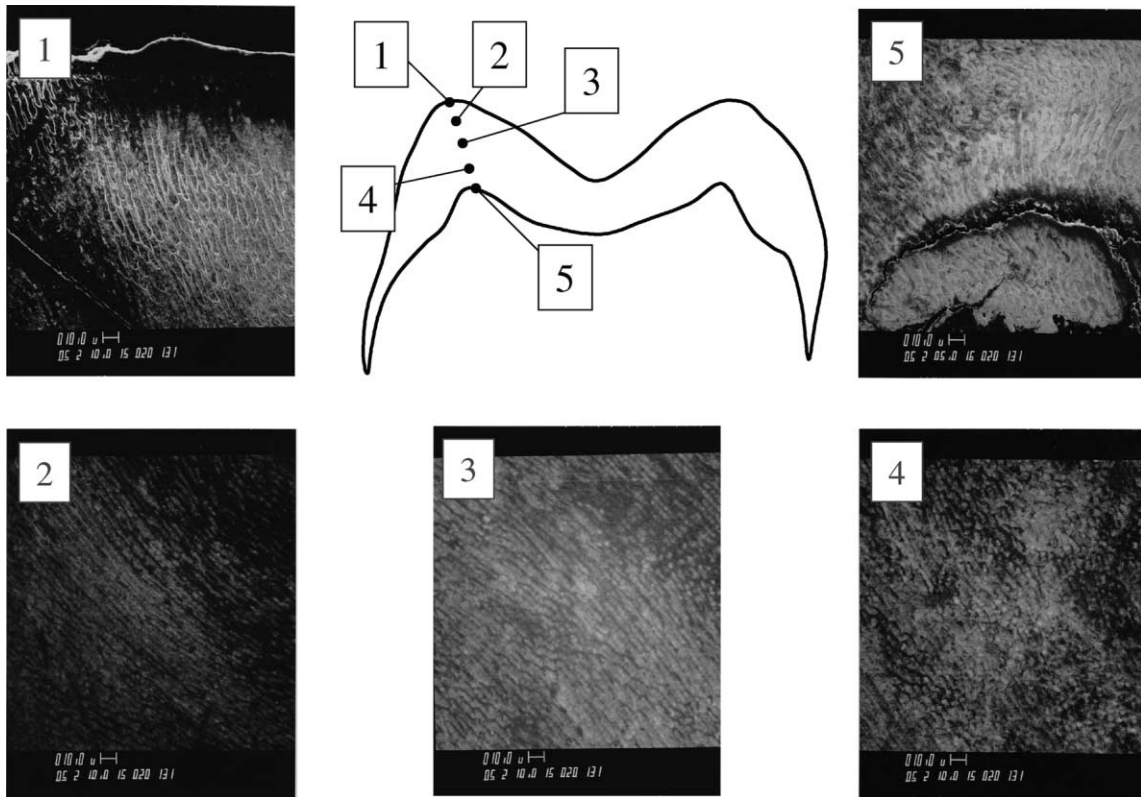


Fig. 5. Buccal traverse: scanning electron microscopic images of the axial section surface of a maxillary M².

obtained in averaging the local values of these properties. The trends reported here are similar to, but larger than, those reported earlier for the hardness of enamel. Using Knoop indentation, Willems et al. (1993) reported 10% decrease in Knoop hardness from 287 for surface enamel to 259 for enamel at the junction. Knoop indentation in that case, however, was initiated some 300 μm from the enamel surface and only extended to 1,500 μm below the surface, thus, the large variations observed here would have been missed. We have shown that the highest values for H and E lie within 100 μm of the occlusal surface and the lowest are in specific regions close to the enamel–dentine junction. In addition, during the earlier Knoop indentation testing by Willems et al. (1993), it was not possible to dependably determine E and, thus, trends in this property could not be established.

Variations in H and E between the buccal and lingual sides of the tooth are also evident in our nanoindentation results. In general, both H and E were higher along the surface of the lingual cusp than along the buccal cusp. In contrast, the interior of the lingual cusp appeared to have a lower H and E than the interior of the buccal cusp.

The mechanical-property maps in Fig. 2, which cover the full axial cross-section of a human molar, demonstrate the variability in the properties of human enamel. While the number of samples examined is limited, and there is clearly

room for further testing, these results, to the best of our knowledge, represent the most detailed analysis of variations in H and E within an individual tooth to date. The comparisons in Fig. 2 between previously reported values for H and E and the current mechanical-property maps indicate the striking difference between the past and present results. Note in Fig. 2 that the past investigations of mechanical properties yielded mainly average values of H and E that correspond to the values for the interior enamel. These previous studies have not attempted, or have not been able, to detect the extreme variations seen here in small areas of the enamel, particularly near the occlusal surface. Only two earlier nanoindentation studies have shown any evidence for the highest E and H found here. Willems et al., 1993 reported $E = 90.59 \pm 16.13$ GPa and Mahoney et al. (2000) reported $H = 4.88 \pm 0.35$ GPa.

Over the last 25 years, many researchers have characterized the chemistry of human enamel as a function of position (Weatherell et al., 1974; Annegarn et al., 1981; Theuns et al., 1983; Svalbe et al., 1984a and b; Wilson and Beynon, 1989), and in some cases the results have been correlated with spatial variations in its mechanical properties (Staines et al., 1981; Svalbe et al., 1984a; Kodaka et al., 1992; Spears, 1997). Here we note that the ranges of Na₂O, MgO, Cl and F concentrations are similar for the centre, buccal, and lingual

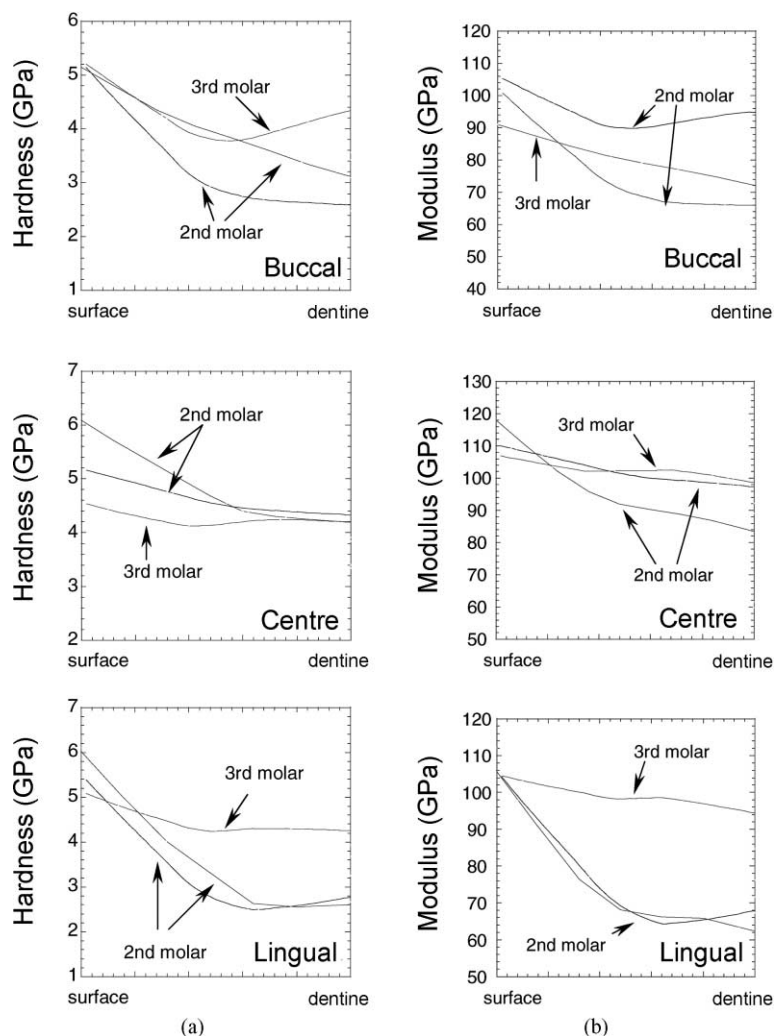


Fig. 6. Comparison between the mechanical properties of different teeth: (a) the H for two different maxillary M² and a M³; (b) the E for two different maxillary M² and a M³.

traverses in Fig. 3, suggesting that variations in the concentration of these elements do not explain the major differences in H and E that were observed for the three regions. The measurements of CaO and P₂O₅ concentrations are more likely sources. The highest CaO or P₂O₅ were found at the enamel surface of the centre traverse, and this location also displayed the highest H and E of the entire molar. The traverse with the lowest CaO or P₂O₅ was the lingual cusp, and this interior region also showed the lowest H and E . These results suggest that hydroxyapatite's mechanical properties depend strongly on its degree of mineralization (CaO/P₂O₅ content), as was suspected previously (Staines et al., 1981; Svalbe et al., 1984a; Spears, 1997). Of course, reductions in mineralization near the enamel–dentine junction have also been associated with increased porosity and increased water

content (Shellis and Dibdin, 2000), which could also act to reduce H and E .

In addition to the general trends in chemistry and their potential impact on mechanical properties, it is also interesting to consider local variations in chemistry that result from microstructural features such as rods, tufts, lamellae and spindles (Iwai-Liao et al., 1997). Enamel tufts, lamellae and spindles contain larger quantities of organic material and are lower in total CaO/P₂O₅ wt.% than surrounding, highly mineralized enamel rods. These structures are also correlated with specific regions within the enamel. Tufts fan out from anchor points near the enamel–dentine junction, lamellae tend to be 'cracks' running the entire width of the enamel, and spindles have been observed at the cuspal summits. These general observations compare favourably

with the trends for CaO and P_2O_5 in (Fig. 3). For example, the lower wt.% of CaO and P_2O_5 near the enamel–dentine junction could be attributed to the presence of hypomineralized tufts near the dentine. Similarly, the isolated fall in Ca/P shown in the lingual-cusp traverse (Fig. 3) could be attributable to the presence of lamellae. Future nanoindentation studies should attempt to characterize the mechanical properties of these local microstructures, but finer spatial resolution will be required.

We should also consider the influence of prism orientation on the mechanical properties of enamel. Our scanning electron micrographs showed some variation in prism alignment with position within the enamel. For the axial cross-sections that were characterized here, the long axes of prisms were aligned perpendicular to the occlusal surface near the surface of the enamel and, thus, were predominantly parallel to the sample's surface. Near the enamel–dentine junction, the long axes of the prisms were generally aligned parallel to the occlusal surface, but they also varied randomly with respect to the surface used for nanoindentation, as they sit both perpendicular and parallel to that surface. Given this alignment of the prisms relative to the sample's surface, the nanoindenter moved perpendicular to the axes of the prisms near the enamel surface (transverse loading), and it moved both parallel and perpendicular to the axes of the prisms at the junction (parallel and transverse loading). Based on the work of Spears, who examined the effects of loading geometry (Spears, 1997), one might expect the present values of H and E to be lower at the enamel surface where the prisms experience just transverse loading than at the enamel–dentine junction where the prisms experience both parallel and transverse loading. However, the opposite trend was observed: H decreased on average by 26% on going from the enamel surface to the dentine, and E decreased on average by 27%. The extreme variations were much larger. This finding suggests that prism alignment or loading direction may not produce the observed trends in H and E .

If prism alignment during indentation had a large impact on the measured H and E , then we would also expect larger S.D.s nearer the enamel–dentine junction because of the increased variation in prism alignment and, hence, loading direction (transverse versus parallel). However, the S.D.s did not show any consistent trends. Thus, prism alignment and loading direction appear to have limited influence on the values of H and E measured here.

While the reported variations in mechanical properties do not follow the predictions made based on the loading geometry of prisms, they do agree with some earlier work. Staines et al. (1981) report that the E of side enamel was approximately 24% greater for areas in which indentation was performed transverse to the prisms (as at the occlusal surface) than for areas in which the indentation was conducted transverse and parallel to the prisms (as at the edge of the dentine). In both this earlier study and the current one, enamel chemistry may be the controlling factor, as suggested by Staines et al. (1981).

As shown in Figs. 4 and 5, the percentage of visible interprismatic regions appeared to be higher in the enamel near the junction with dentine than in the surface enamel. A higher percentage of presumably organic/hypomineralized material corresponds to a lower degree of mineralization, and this correlates with the lower calcium and phosphate contents that were observed near that junction. As hypomineralized tissue is more common in the interior layers of enamel, and probably softer and more compliant, as well, the interior layers of enamel might be expected to be softer and more compliant, regardless of indenter/prism loading geometry. However, given the complex decussation of prisms in human enamel, more precise monitoring of prism orientation, microstructure, and degree of mineralization is needed to evaluate definitively the relative influence of mineralization (chemistry) and prism orientation on the mechanical properties of enamel.

Based on the sharp spatial variations in mechanical properties seen in Figs. 2 and 6, future studies that characterize the mechanical properties of enamel in a group of teeth must take into account the large intra-tooth variations that have been observed here. Specifically, test sites must be chosen carefully so that intra-tooth variations do not dwarf the differences observed between teeth due to factors such as type, history or age of the tooth. Furthermore, such detailed mapping of the mechanical, chemical and structural properties of teeth provides an opportunity to explore the relation between form and function for human enamel for molars, canines and incisors, for maxillary and mandibular teeth, and for deciduous and permanent teeth.

Consider the variations in H and E between the buccal and lingual sides of the tooth shown in Fig. 2. In general, both H and E are higher along the surface of the lingual cusp than along the buccal cusp. In contrast, the interior of the lingual cusp appears to have a lower H and E than the interior of the buccal cusp. The differences between the two halves of the tooth may ultimately be tied to the functional demands placed on the tooth during mastication. The lingual cusp is known to experience large loads acting normal to its occlusal surface during mastication, especially as one moves posteriorly along the tooth row (Spears and Macho, 1998). Thus, H and E may be higher at this surface (Fig. 2) to shield the underlying enamel and dentine. In the intercuspal region, the applied loads are large at the surface as well. But, the loads are also thought to be large below the intercuspal surface because a bolus of food can apply a significant bending moment to this region. Thus, we would expect H and E to be high throughout the intercuspal region as observed. By contrast, the buccal cusp of the maxillary molar typically experiences lower loads during mastication and is less likely to have a hard and stiff outer surface that shields the underlying enamel and dentine, as seen in Fig. 2.

These hypotheses linking variations in mechanical properties and tooth function follow earlier arguments on enamel thickness. Enamel on the lingual cusps of maxillary molars is generally thicker than that on the buccal cusps (Khera

et al., 1990; Macho and Berner, 1994; Schwartz, 2000). Finite-element analyses suggest that this enhanced thickness may lower tensile stresses in the intercusp regions when the lingual cusp is loaded (Macho and Spears, 1999). Our present results allow us to refine this suggestion. In other words, the changes in E and H as one moves from the surface of the lingual cusp to the enamel–dentine junction, in conjunction with the thickness of the enamel, could result in a stress-shielding mechanism for the interior portions of the cusp. The same could hold true for the intercusp area, where we found the highest measures of H and E , and finite-element studies report the highest stresses due to mastication (Spears and Macho, 1998).

The results for other molar samples (Fig. 6) show that the H and E of molar enamel are generally higher at the surface than at the enamel–dentine junction. Moreover, M^3 tends to show similar surface properties to M^2 . However, the interior regions below the buccal and lingual cusps of M^3 are harder and stiffer than the corresponding regions in M^2 , particularly for the lingual cusp. Once again, these results may ultimately be tied to functional demands. Specifically, as M^3 is closer to the temporomandibular joint and more buccally tilted, one might expect higher occlusal loads than on M^1 or M^2 , especially on the lingual cusp (Koolstra et al., 1988; Mansour and Reynick, 1975; Spears and Macho, 1998). The higher interior stiffness H and E on M^3 would seem to be an effective means of coping with these higher loads. With the ability to map variations in mechanical properties in detail, one can now explore these suggested links between form and function using larger sets of samples and a variety of tooth types.

5. Conclusions

The mechanical properties, chemistry and microstructure of enamel in a maxillary M^2 were characterized as a function of position on an axial cross-section. The resulting maps of H and E show striking variations in these properties with distinct decreases on traversing from the occlusal surface to the enamel–dentine junction. At the enamel surface $H > 6$ GPa and $E > 115$ GPa, while at the enamel–dentine junction $H < 3$ GPa and $E < 70$ GPa. These significant decreases show a strong correlation with reductions in the wt.% of P_2O_5 and CaO and increases in the wt.% of Na_2O and MgO. They also show a weak correlation with the degree of prism alignment. Taken together, these results support the hypothesis that the mechanical properties of enamel are strongly dependent on its degree of mineralization (CaO/ P_2O_5 content) and only weakly dependent on its microstructure.

Besides distinct declines in H and E on moving away from the occlusal surface, the mechanical properties of the enamel also differ from the lingual to the buccal side of the molar. The lingual side, which experiences higher surface loads during mastication, has higher values of H and E at its occlusal surface and lower values at the enamel–dentine

junction than the buccal side. These differences, together with the generally higher H and E of M^3 , suggest that tooth function may be driving a non-uniform distribution of properties within human enamel.

The significant variations reported here for the H and E of enamel strongly suggest that future comparative studies, aimed at assessing differences in the mechanical properties of enamel, must be careful to account for intra-tooth position. Without such care, the differences within a population of teeth, whether that population varies by type, age or sex, may be completely overwhelmed by variations due to intra-tooth position.

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References

- Annegarn, H.J., Jodaikin, A., Cleaton-Jones, P.E., Sellschop, J.P.F., Madiba, C.C.P., Bibby, D., 1981. PIXE analysis of caries related trace elements in tooth enamel. *Nucl. Instr. Methods* 181, 323–326.
- Bodart, F., Deconninck, G., Martin, M.T., 1981. Large scale study of tooth enamel. *IEEE Trans. Nucl. Sci.* NS-28 (2), 1401–1403.
- Craig, R.G., Peyton, F.A., 1958. The microhardness of enamel and dentin. *J. Dent. Res.* 37 (4), 661–668.
- Craig, R.G., Peyton, F.A., Johnson, D.W., 1961. Compressive properties of enamel, dental cements, and gold. *J. Dent. Res.* 40 (5), 936–945.
- Iwai-Liao, Y., Higashi, Y., Guo, L., Ogita, Y., Nonaka, T., 1997. Fine structure study and energy dispersive spectroscopy (EDS) X-ray microprobe (EDX) analysis of organic elements in the human enamel. *J. Trace Microprob. Tech.* 15 (4), 673–682.
- Khera, S.C., Carpenter, C.W., Vetter, J.D., Staley, R.N., 1990. Anatomy of cusps of posterior teeth and their fracture potential. *J. Prost. Dent.* 64, 139–147.
- Kinney, J.H., Balooch, M., Marshall, S.J., Marshall Jr., G.W., Weihs, T.P., 1996. Hardness and Young's modulus of human peritubular and intertubular dentine. *Archs. Oral Biol.* 41 (1), 9–13.
- Kodaka, T., Debari, K., Yamada, M., Kuroiwa, M., 1992. Correlation between microhardness and mineral content in sound human enamel. *Caries Res.* 26, 139–141.
- Koolstra, J.H., van Eijden, T.M.J.G., Weijs, W.A., Naeije, M., 1988. A three-dimensional mathematical model of the human

- masticatory system predicting maximum bite forces. *J. Biomech.* 21, 563–576.
- Macho, G.A., Berner, M.E., 1994. Enamel thickness and the helicoidal occlusal plane. *Am. J. Phys. Anthropol.* 94, 327–337.
- Macho, G.A., Spears, I.R., 1999. Effects of loading on the biomechanical behavior of molars of *Homo*, *Pan*, and *Pongo*. *Am. J. Phys. Anthropol.* 109, 211–227.
- Mahoney, E., Holt, A., Swain, M., Kilpatrick, N., 2000. The hardness and modulus of elasticity of primary molar teeth: an ultra-micro-indentation study. *J. Dent.* 28, 589–594.
- Mansour, R.M., Reynick, R.J., 1975. In vivo occlusal forces and moments. Part I. Forces measured in terminal hinge position and associated movements. *J. Dent. Res.* 54, 114–120.
- Meredith, N., Sherriff, M., Setchell, D.J., Swanson, S.A.V., 1996. Measurement of the microhardness and Young's modulus of human enamel and dentine using an indentation technique. *Archs. Oral Biol.* 41 (6), 539–545.
- Oliver, W.C., Pharr, G.M., 1992. An improved technique for determining hardness and Young's modulus using load and displacement sensing indentation experiments. *J. Mater. Res.* 7 (6), 1564–1583.
- Rasmussen, S.T., Patchin, R.E., Scott, D.B., Heuer, A.H., 1976. Fracture properties of human enamel and dentin. *J. Dent. Res.* 55 (1), 154–164.
- Remizov, S.M., Prujansky, L.Y., Matveevsky, R.M., 1991. Wear resistance and microhardness of human teeth. *Proc. Instr. Mech. Eng. Part H J. Eng. Med.* 205 (3), 201–202.
- Schwartz, G.T., 2000. Taxonomic and functional aspects of the patterning of enamel thickness distribution in extant large-bodied hominoids. *Am. J. Phys. Anthropol.* 111, 221–244.
- Shellis, R.P., Dibdin, G.H., 2000. Enamel microporosity and its functional implications. In: Teaford, M.F., Smith, M.M., Ferguson, M.W.J. (Eds.), *Development, Function and Evolution of Teeth*, Cambridge University Press, Cambridge, pp. 242–251.
- Spears, I.R., 1997. A three-dimensional finite element model of prismatic enamel: a re-appraisal of the data on the Young's modulus of enamel. *J. Dent. Res.* 76 (10), 1690–1697.
- Spears, I.R., Macho, G.A., 1998. Biomechanical behavior of modern human molars: implications for interpreting the fossil record. *Am. J. Phys. Anthropol.* 106, 467–482.
- Staines, M., Robinson, W.H., Hood, J.A.A., 1981. Spherical indentation of tooth enamel. *J. Mater. Sci.* 16, 2551–2556.
- Svalbe, I.D., Chaudhri, M.A., Traxel, K., Ender, C., Mandel, A., 1984a. Microprobe profiling of fluorine and other trace elements to large depths in teeth. *Nucl. Instr. Method Phys. Res. (Pt B3)*:648–650.
- Svalbe, I.D., Chaudhri, M.A., Traxel, K., Ender, C., Mandel, A., 1984b. Surface profiling of trace elements across pre-carious lesions in teeth. *Nucl. Instr. Method Phys. Res. (Pt B3)*:651–653.
- Theuns, H.M., van Dijk, J.W.E., Jongebloed, W.L., Groeneveld, A., 1983. The mineral content of human enamel studied by polarizing microscopy, microradiography and scanning electron microscopy. *Archs. Oral Biol.* 28, 797–803.
- van Meerbeek, B., Willems, G., Celis, J.P., Roos, J.R., Braem, M., Lambrechts, P., Vanherle, G., 1993. Assessment by nano-indentation of the hardness and elasticity of the resin–dentin bonding area. *J. Dent. Res.* 72 (10), 1434–1442.
- Weatherell, J.A., Robinson, C., Hallsworth, A.S., 1974. Variations in the chemical composition of human enamel. *J. Dent. Res.* 53, 180–192.
- Willems, G., Celis, J.P., Lambrechts, P., Braem, M., Vanherle, G., 1993. Hardness and Young's modulus determined by nanoindentation technique of filler particles of dental restorative materials compared with human enamel. *J. Biomed. Mater. Res.* 27, 747–755.
- Williams, P.L., Warwick, R., Dyson, M., Bannister, L.H. (Eds.), 1989. *Splanchnology: The teeth*. In: Gray's Anatomy, Churchill Livingstone, New York, pp. 1308–1309.
- Wilson, P.R., Beynon, A.D., 1989. Mineralization differences between human deciduous and permanent enamel measured by quantitative microradiography. *Arch. Oral Biol.* 34, 85–88.
- Xu, H.H.K., Smith, D.T., Jahanmir, S., Romberg, E., Kelly, J.R., Thompson, V.P., Rekow, E.D., 1998. Indentation damage and mechanical properties of human enamel and dentin. *J. Dent. Res.* 77 (3), 472–480.